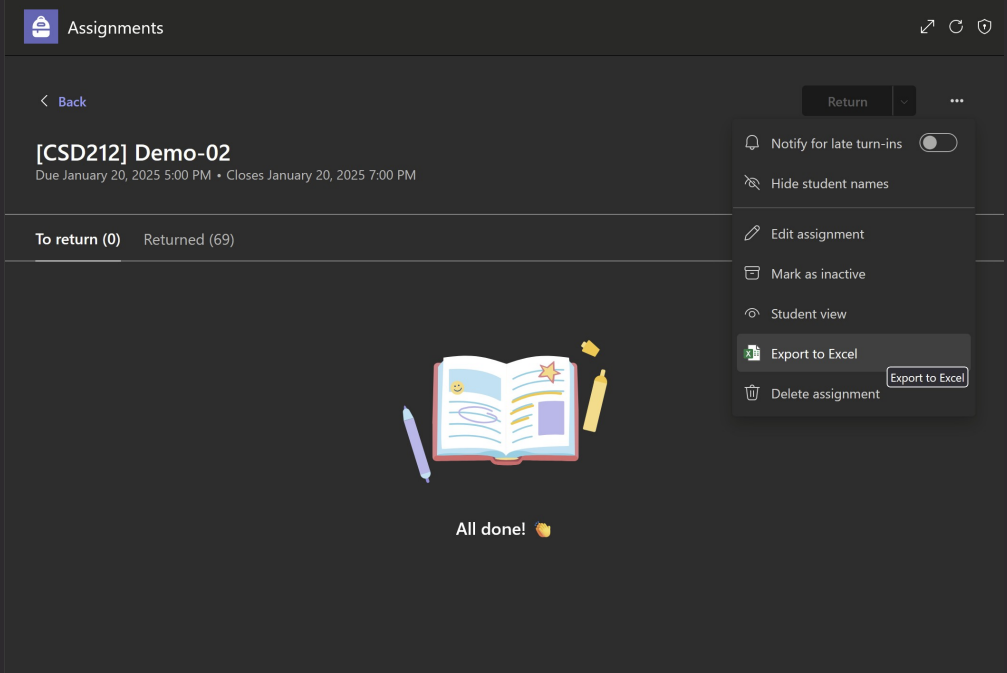


Detector (`detector.py`)

These are the parameters, input and output for the module. Kindly note that this is to be run individually for each demo.

Required Input

1. Ensure that preprocessed student assignments are available in the specified folder for the required demo.
2. Export demo grades as Excel file from Teams after returning all of them.



Expected Output

1. Generates an Excel report with plagiarism scores for each student and their corresponding calculated final grades.
2. Final grades are calculated by penalizing the student's grade in the exported Excel if the plagiarism is above acceptable thresholds.

student_name	max_similarity	similarity_with	plagiarism_penalized	marks_given	final_marks
A J	1	D N	TRUE	8	0
A G	0.34470377	A M	FALSE	0	0
A Y	1	P A	TRUE	7	0
A B	0.989528796	D P	TRUE	10	5
A S	0.430278884	S M	FALSE	4	4
A A	0.558752998	N M	FALSE	10	10
A G	0.677884615	G B	FALSE	8.5	8.5
A P	0.303135889	A J	FALSE	5	5
A G	0.728	N M	FALSE	8.5	8.5
A M	0.578125	S S	FALSE	10	10

Parameters

paths

- `demo` *(int)*
The demo number for which plagiarism detection is being run.
- `preprocessed_assignments_folder` *(str)*
Directory where formatted assignments are stored after preprocessing.
- `plagiarism_excel_save_path` *(str)*
Path where the plagiarism detection results will be saved as an Excel file.
- `teams_excel_path` *(str)*
Path to the exported Teams Excel file containing student grades.

fingerprint

- `k` *(int)*
Length of k-grams to extract as fingerprints.
- `win_size` *(int)*
Window size to use for winnowing (must be >= 1).
- `boilerplate` *(list(str))*
A list of common lines (e.g., matplotlib calls) and library imports that should be ignored during plagiarism detection.

plagiarism

- `penalties` *(dict[float, float])*
A mapping of plagiarism thresholds to the penalties applied.
(e.g. {0.8: 0.25, 0.9: 0.5} means if plagiarism >= 0.9, 50% of marks are deducted and if 0.9 > plagiarism >= 0.8, 25% of marks are deducted)

excel

- `teams_row_start` *(int)*
The starting row after headers in the Teams Excel file.
- `teams_name_col` *(str)*
The column in the Teams Excel file containing student names.
- `teams_marks_col` *(str)*
The column in the Teams Excel file containing student marks.

Configuration

All parameters, including demo mappings, boilerplate exclusions, and detection thresholds, are stored in a YAML file (`config.yaml`). Users can modify this file instead of changing code directly.

Contact

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Developed for internal use in CSD212 at Shiv Nadar University.
If you have any questions or suggestions for improvement, feel free to reach out.